

Soil Sustainability Score Calculator

1. Introduction

Soil is a critical resource for agriculture and environmental sustainability. Factors such as nutrient depletion, salinity, erosion, and declining organic matter reduce soil productivity over time. The Soil Sustainability Score Calculator aims to assess soil health and generate a sustainability score to support farmers, agriculture officers, and government agencies in making informed decisions.

2. Problem Statement

- Existing soil test reports provide individual soil parameters but do not indicate overall soil sustainability.
- Soil degradation due to nutrient depletion, salinity, erosion, poor soil texture, and low water holding capacity affects agricultural productivity.
- Farmers and agriculture officers face difficulties in interpreting soil health data and identifying sustainability risks.
- There is a need for a centralized system that evaluates soil health and generates a Soil Sustainability Score for sustainable soil management and decision-making.

3. Objectives

- Assess soil health using scientific indicators.
- Generate a Soil Sustainability Score.
- Classify soil sustainability levels.
- Provide recommendations for soil improvement.
- Support soil monitoring through GIS-based visualization.

4. Users

Farmer

- View soil sustainability score.
- Access soil health reports.
- Receive recommendations.

Agriculture Officer

- Upload and verify soil records.
- Monitor village-level soil health.
- Generate reports.

Government Administrator

- Monitor district-level sustainability.
- Analyze trends and generate policy reports.

5. Data Sources

- Soil Health Card Data
- Soil Testing Laboratory Reports
- Land Records
- Weather Data (Optional)

Parameters Used

- pH
- Organic Carbon (OC)
- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)
- Electrical Conductivity (EC)
- Soil Texture

6. Methodology

1. Collect soil data.
2. Validate and preprocess data.
3. Calculate soil health indicators.
4. Apply machine learning models.
5. Generate Soil Sustainability Score.
6. Provide recommendations.
7. Visualize results on GIS maps.

7. Machine Learning Methods

Random Forest

- Handles structured soil data effectively.
- Used for soil sustainability classification.

XGBoost

- Provides high prediction accuracy.
- Used for sustainability score prediction.

CNN (Future Scope)

- Suitable for satellite image analysis.
- Can be used for erosion and soil degradation detection.

8. Scoring Methodology

Chemical Health (70%)

Physical Health (30%)

Formula

Soil Sustainability Score (SSS)

$$\text{SSS} = (0.70 \times \text{Chemical Health Score}) + (0.30 \times \text{Physical Health Score})$$

Categories

Score	Category
80–100	Highly Sustainable
60–79	Moderately Sustainable
40–59	Needs Improvement
Below 40	Unsustainable

9. Flowchart

Start

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Collect Soil Data

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Data Validation & Preprocessing

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Random Forest / XGBoost Analysis

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Calculate Soil Sustainability Score

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Generate Recommendations

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Store Results

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Display Dashboard & GIS Maps

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End

10. Technologies Used

Frontend : HTML , CSS, JavaScript, Bootstrap

Backend: Flask

Database : PostgreSQL, PostGIS

Programming & Analytics: Python, Pandas ,NumPy

Machine Learning : Random Forest , XGBoost

GIS : QGIS , GeoPandas

Visualization: Plotly, Matplotlib

11. Expected Outputs

- Soil Sustainability Score
- Sustainability Category
- Soil Health Report
- Improvement Recommendations
- GIS-Based Soil Maps

12. Conclusion

The Soil Sustainability Score Calculator provides a scientific and data-driven approach to evaluate soil health and sustainability. By combining soil parameters, machine learning, and GIS visualization, the system helps improve soil management and supports sustainable agricultural development.